

Program : Diploma in Biomedical Engineering	
Course Code : 6241C	Course Title: Bio MEMS
Semester : 6	Credits: 4
Course Category : Program Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester : 60

Course Objectives:

- To introduce various MEMS fabrication techniques
- To impart knowledge on different types of sensors and actuators and their principles of operation at the micro scale level.
- To give an overview of the application of MEMS in different field of medicine.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Describe the MEMS materials and illustrate fabrication process of MEMS devices	18	Understanding
CO2	Enumerate and explain the MEMS sensors and Actuators and conduct a survey on MEMS sensors available in market.	14	Applying
CO3	Classify and explain the functioning of microfluidic devices used in MEMS	14	Understanding
CO4	Explain the various applications of BioMEMS and make a presentation on Lab-on-a-chip	12	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3					3	3
CO3	2						
CO4	2					3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the MEMS materials and illustrate the fabrication process of MEMS devices		
M1.01	Summarize typical Micro electro mechanical system	5	Understanding
M1.02	Explain the materials used for MEMS fabrication and its properties	6	Understanding
M1.03	Illustrate the various fabrication processes of MEMS	7	Understanding
Contents: MEMS MATERIALS AND FABRICATION Typical MEMS and Microsystems, materials for MEMS - active substrate materials- Silicon and its compounds, Silicon piezoresistors, Gallium Arsenide, quartz, polymers. Micromachining photolithography, thin film deposition, doping, etching, bulk micromachining, wafer bonding, LIGA.			
CO2	Enumerate and explain the MEMS sensors and Actuators and conduct a survey on MEMS sensors available in market.		
M2.01	Enumerate and describe the mechanical sensors and actuators used in MEMS	4	Understanding
M2.02	Enumerate and describe the thermal sensors and actuators used in MEMS	4	Understanding
M2.03	Enumerate and describe the electrostatic sensors and actuators used in MEMS	4	Understanding
M2.04	Perform a survey of MEMS sensors available in the market.	2	Applying
	Series Test I	1	

Contents:**MEMS SENSORS AND ACTUATORS**

Mechanical sensors and actuators - beam and cantilever - microplates, strain, pressure and flow measurements,

Thermal sensors and actuators - actuator based on thermal expansion, thermocouples, thermal resistor, Shape memory alloys - Inertia sensor, flow sensor

Electrostatic sensors and actuators - Inertia sensor, Pressure sensor, flow sensor, tactile sensor, comb drive

Case study: Perform a survey of MEMS sensors available in market which should include at least five Bio MEMS sensors.

CO3	Classify and explain the functioning of microfluidic devices used in MEMS		
M3.01	Illustrate the parts of micropumps	4	Understanding
M3.02	Compare and contrast various mechanical and non-mechanical micropumps.	5	Understanding
M3.04	Explain the types, shapes, materials and fabrication methods of microneedles	5	Understanding

Contents:**MEMS BASED MICROFLUIDIC DEVICES**

Micropumps- parts of micropumps - actuator - valves- reservoir - Nozzles

Types of micropumps - Mechanical micropumps - Piezoelectric - Electrostatic - Thermopneumatic - Electromagnetic - Bimetallic - Shape Memory Alloy (SMA) micropumps ,

Non-Mechanical Micropumps- Electroosmotic (EO) - Electrowetting (EW) - Electrochemical - Bubble micropumps

Microneedles - Types - shapes - materials - fabrication - applications

CO4	Explain the various applications of BioMEMS		
M4.01	Explain the drug delivery system based on MEMS technology	4	Understanding
M4.02	Explain the polymeric chain reaction and DNA sensors	3	Understanding
M4.03	Illustrate biosensors for glucose, uric acid, urea and triglyceride	4	Understanding
M4.04	Make a presentation on Lab-on-a-chip.	1	Understanding
	Series Test II	1	

Contents:**APPLICATIONS OF BioMEMS**

Introduction to CAD for MEMS, Drug delivery, micro total analysis systems (MicroTAS) detection and measurement methods, microsystem approaches to polymerase chain reaction (PCR), DNA sensor, MEMS based drug delivery, Biosensors- sensors for glucose, uric acid, urea and triglyceride.

Case study: Make a presentation on Lab-on-a-chip.

Text / Reference:

T/R	Book Title/Author
T1	Tai Ran Hsu, - <i>MEMS and Microsystems Design and Manufacture</i> , Tata McGraw Hill Publishing Company, New Delhi, 2002.
T2	WanJun Wang, Stephen A.Soper, <i>BioMEMS: Technologies and Applications</i> , CRC Press, New York, 2007
R1	Muhammad Waseem Ashraf, Shahzadi Tayyaba and Nitin Afzulpurkar, <i>Micro Electromechanical Systems (MEMS) Based Microfluidic Devices for Biomedical Applications</i> , ISSN 1422-0067, International Journal of Molecular Sciences, June 2011
R2	Marc J. Madou - <i>Fundamentals of Microfabrication: the Science of Miniaturization</i> , CRC Press, 2002.

Online resources:

Sl. No	Website Link
1	https://www.youtube.com/watch?v=tOg1CILF2Gk
2	https://nptel.ac.in/courses/112104181/
3	https://www.youtube.com/watch?v=oI0Hgo_dmsg
4	http://www.infocobuild.com/education/audio-video-courses/electronics/FabricationTechniquesMemsBasedSensors-IISc-Bangalore/lecture-46.html
5	https://www.youtube.com/watch?v=eqZgxR6eRjo